

Gabor Vector Fields: Pitch Kernel Smoother Comparison

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May 22, 2026

1 Model

Each Gabor observation supplies $z_i = (x_i, y_i)$ and local pitch α'_i . The global line-field angle is

$$\phi_i = \text{atan2}(y_i, x_i) + \alpha'_i.$$

The smoother is applied to pitch, not directly to the global angle:

$$a_i = (\cos 2\alpha'_i, \sin 2\alpha'_i).$$

For target z with radius r and polar angle ϑ , the fitted pitch embedding is

$$\hat{a}(z) = \frac{\sum_i K(z, z_i) a_i}{\sum_i K(z, z_i)}, \quad \hat{\alpha}'(z) = \frac{1}{2} \text{atan2}(\hat{a}_2(z), \hat{a}_1(z)).$$

The fitted streamline direction is then

$$\hat{\phi}(z) = \text{atan2}(y, x) + \hat{\alpha}'(z).$$

The RBF pitch kernel is

$$K_{\text{RBF}}(z, z'; h) = \exp \left[-\frac{(r - r')^2}{2h^2} \right],$$

and the multiplicative pitch kernel is

$$K_{\text{mult}}(z, z'; h, \kappa) = \exp \left[-\frac{(r - r')^2}{2h^2} \right] \exp [\kappa \cos\{\vartheta - \vartheta'\}].$$

2 Hyperparameter Selection

The fixed bandwidth grid is $h \in \text{geospace}(1, 1500, 72)$. The multiplicative kernel additionally sweeps

$$\kappa \in \{0, 0.125, 0.25, 0.5, 0.75, 1, 1.25, 1.5, 2, 3, 4, 6, 8, 12, 16, 24, 32, 48, 64\}.$$

Each candidate is selected by shuffled 10-fold held-out axial reconstruction RSS. This is used as an anti-overfitting check while keeping the scientific target as streamline description.

3 Results

Model	Mean RSS	Median RSS	Mean MAE deg	h med	κ med
RBF-von-Mises pitch kernel smoother	0.3615	0.3483	26.2678	33.1853	28
RBF pitch kernel smoother	0.5526	0.5822	34.7341	47.6528	

Table 1: Selected pitch-kernel smoother performance across the Gabor vector-field exports.

Model	Mean RSS	Median RSS	Mean MAE deg
RBF-von-Mises pitch kernel smoother	0.3615	0.3483	26.2678
RBF pitch kernel smoother	0.5526	0.5822	34.7341
Parametric continuous p	0.5588	0.5885	35.2181
Parametric p=1 (Archimedean)	0.5949	0.5892	36.668
Parametric p=0 (Logarithmic)	0.6362	0.6331	38.2387
Parametric p=2 (Fermat)	0.6567	0.6875	38.8822

Table 2: Held-out axial reconstruction comparison including the parametric spiral families.

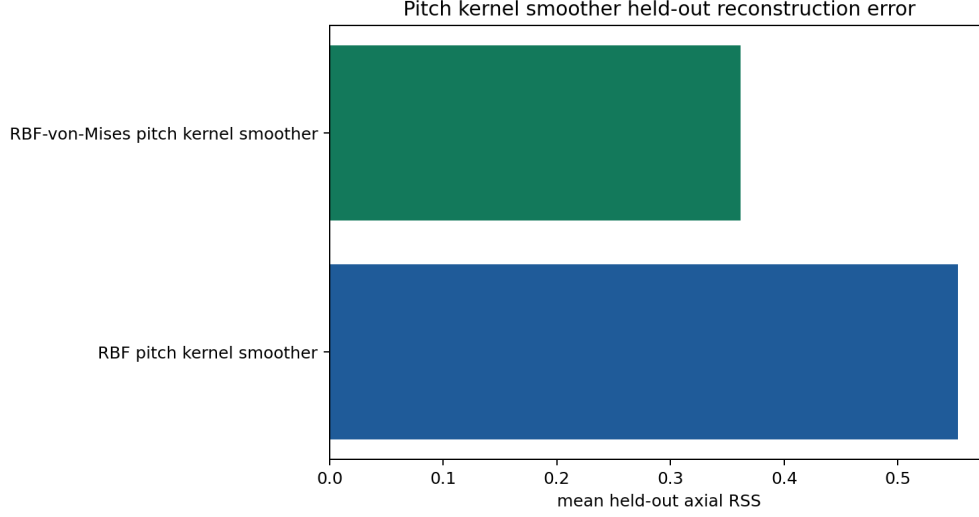


Figure 1: Mean held-out axial RSS for the two pitch-kernel smoothers.

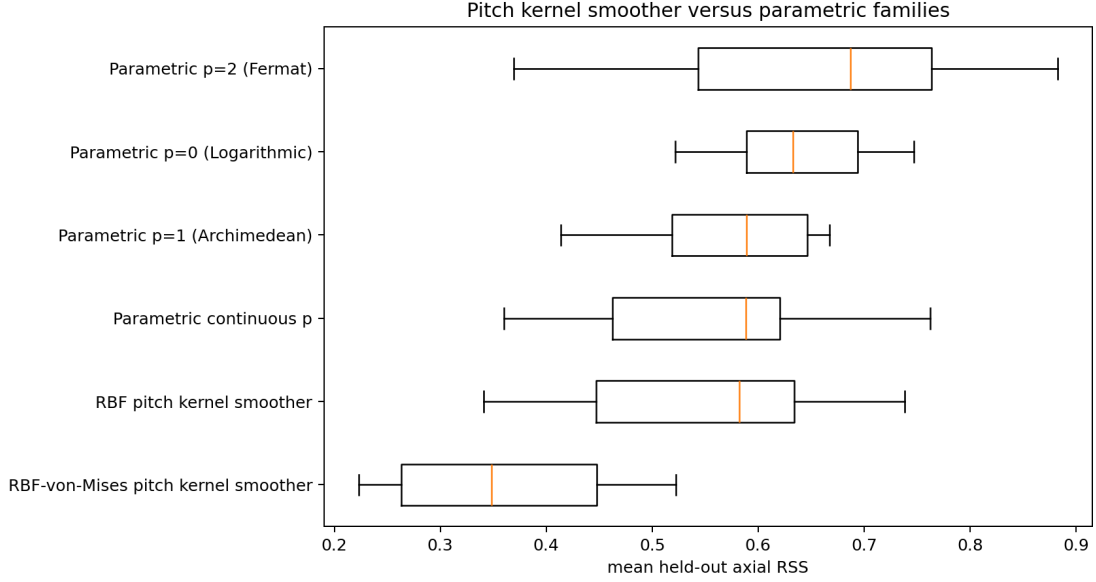


Figure 2: Held-out axial RSS distribution comparing selected pitch-kernel smoothers with parametric families.

4 Streamline Overlays

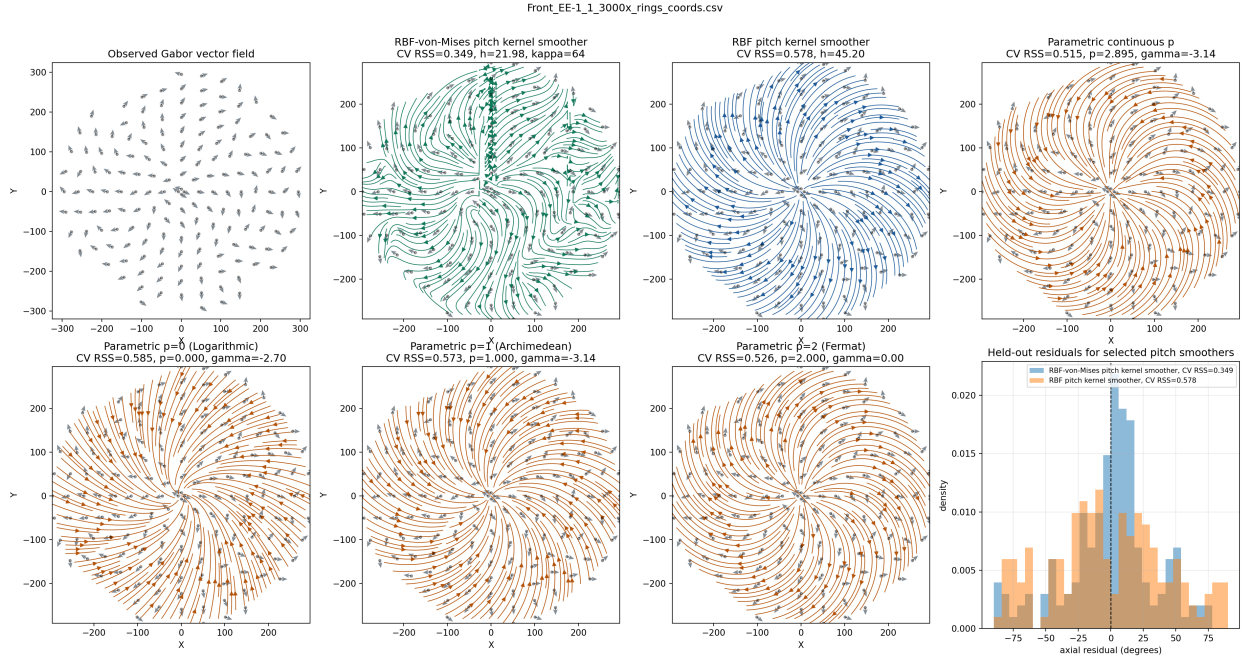


Figure 3: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE-1_1_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

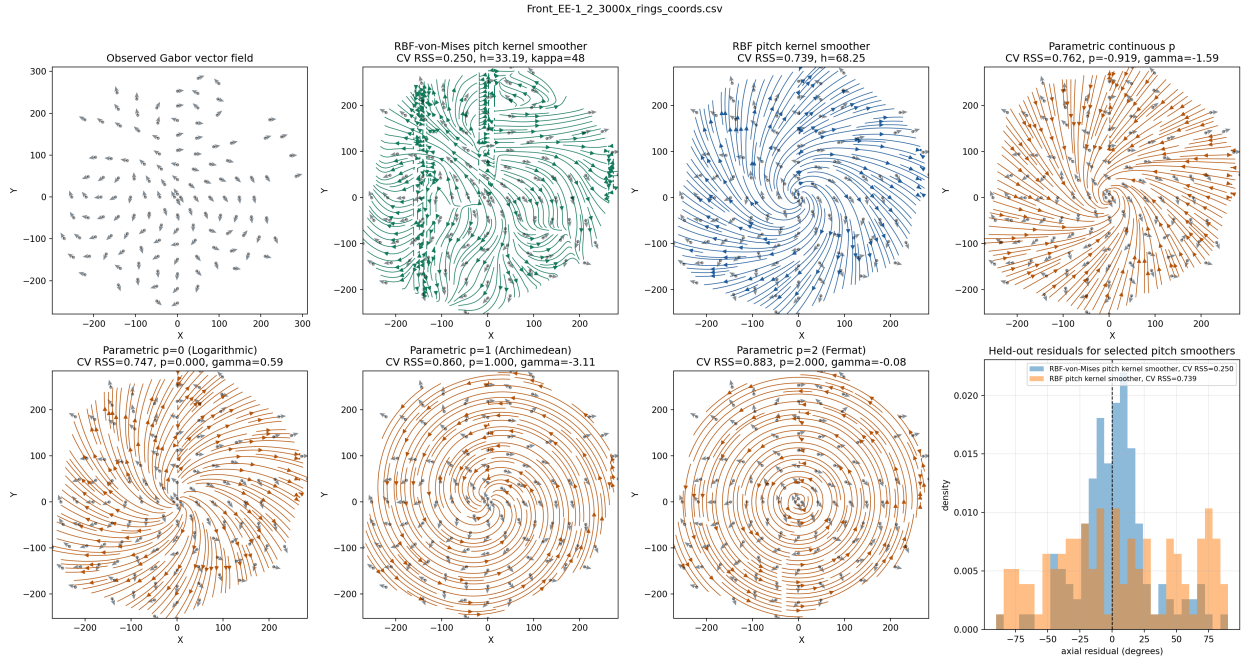


Figure 4: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE-1_2_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

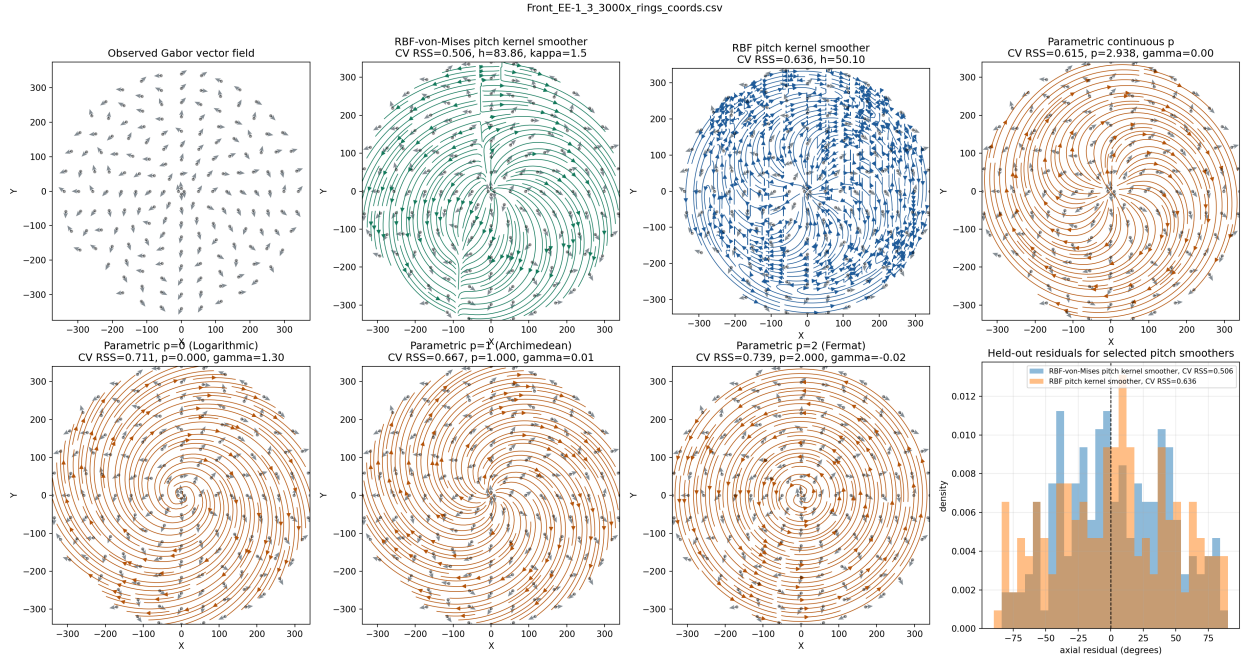


Figure 5: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE-1_3_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

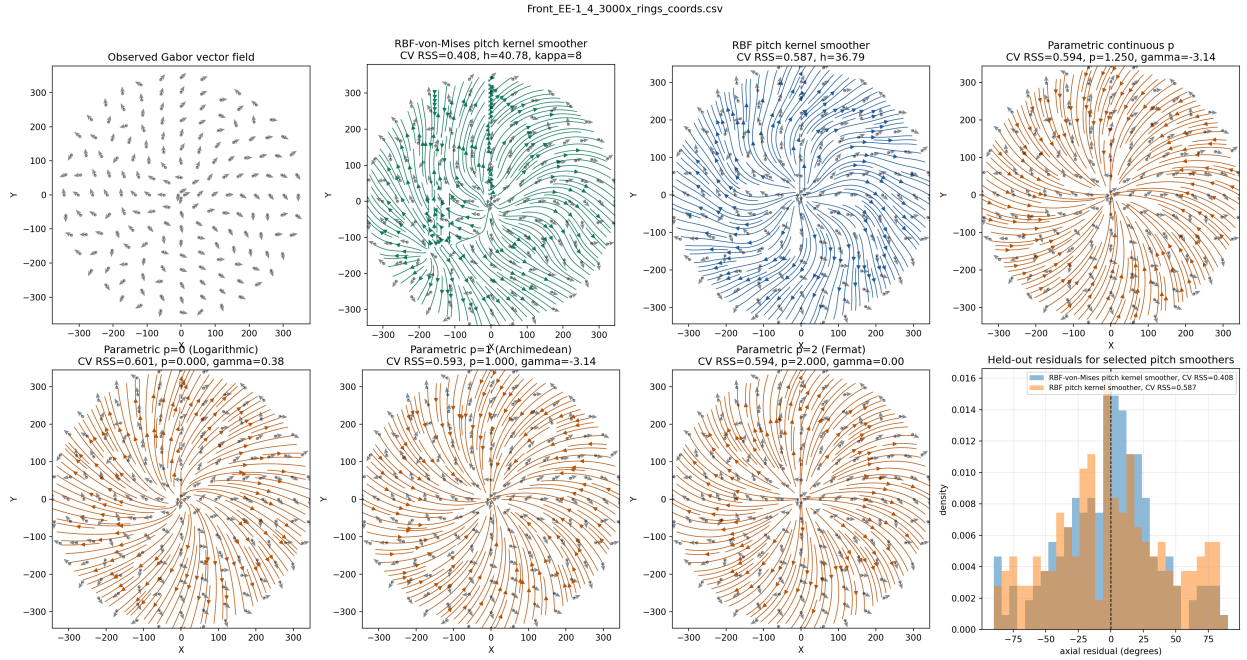


Figure 6: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE-1_4_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

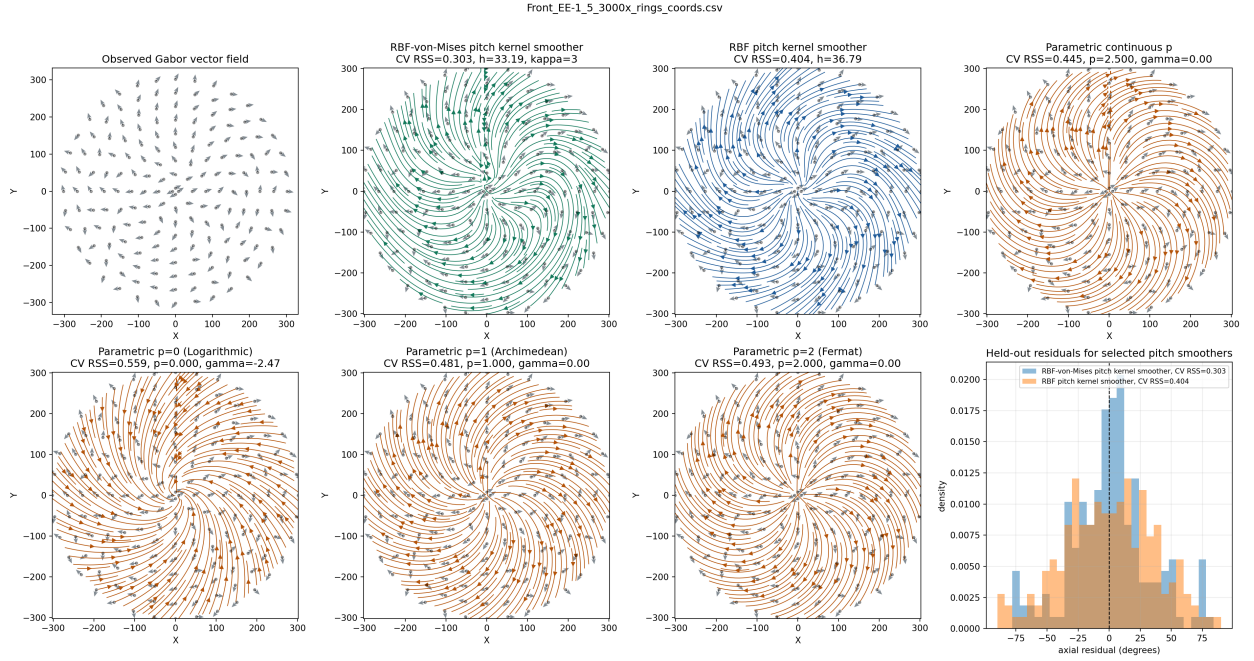


Figure 7: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE-1_5_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

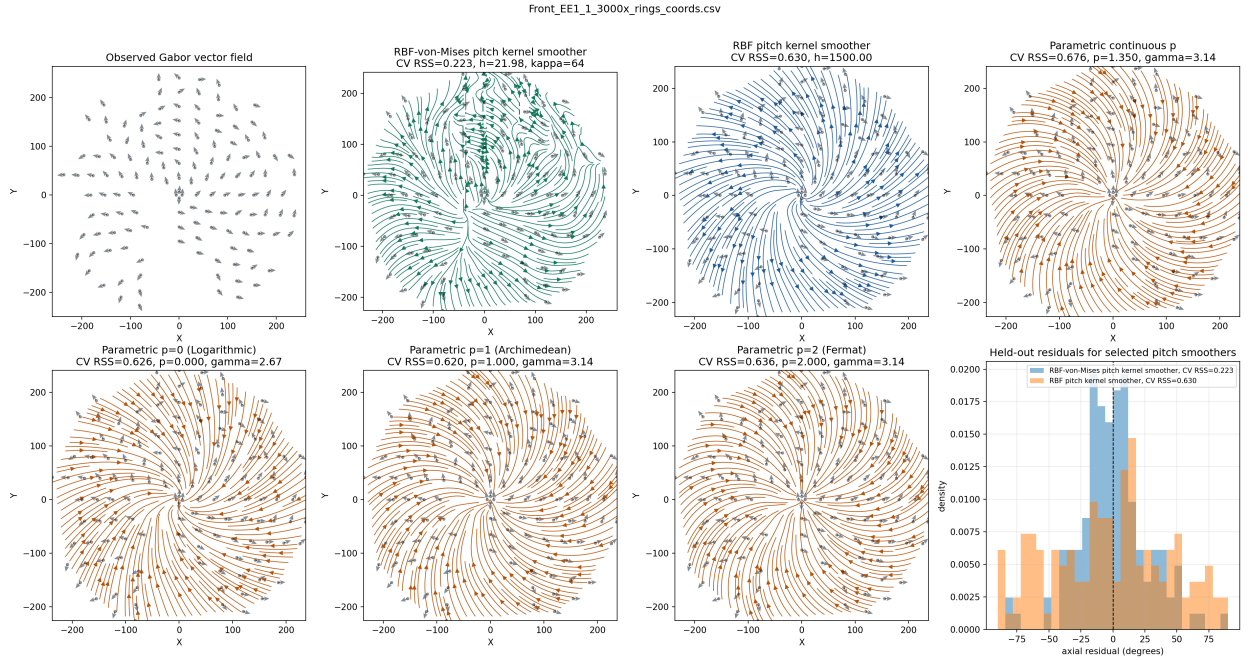


Figure 8: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE1_1_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

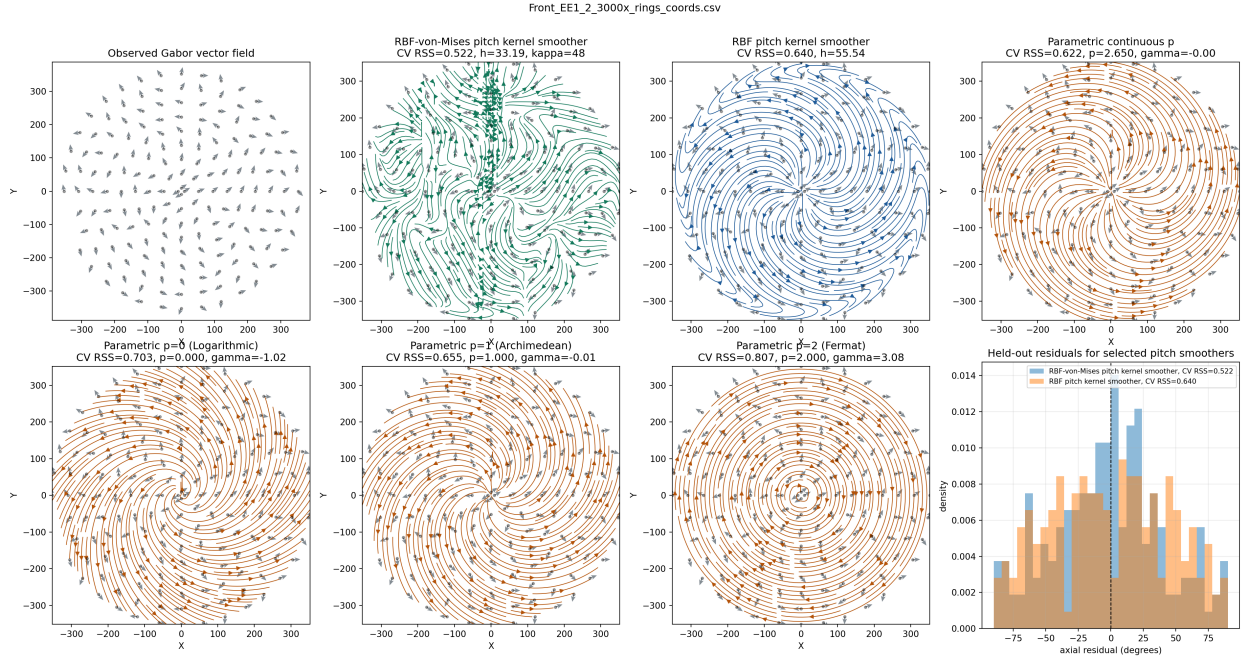


Figure 9: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE1_2_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

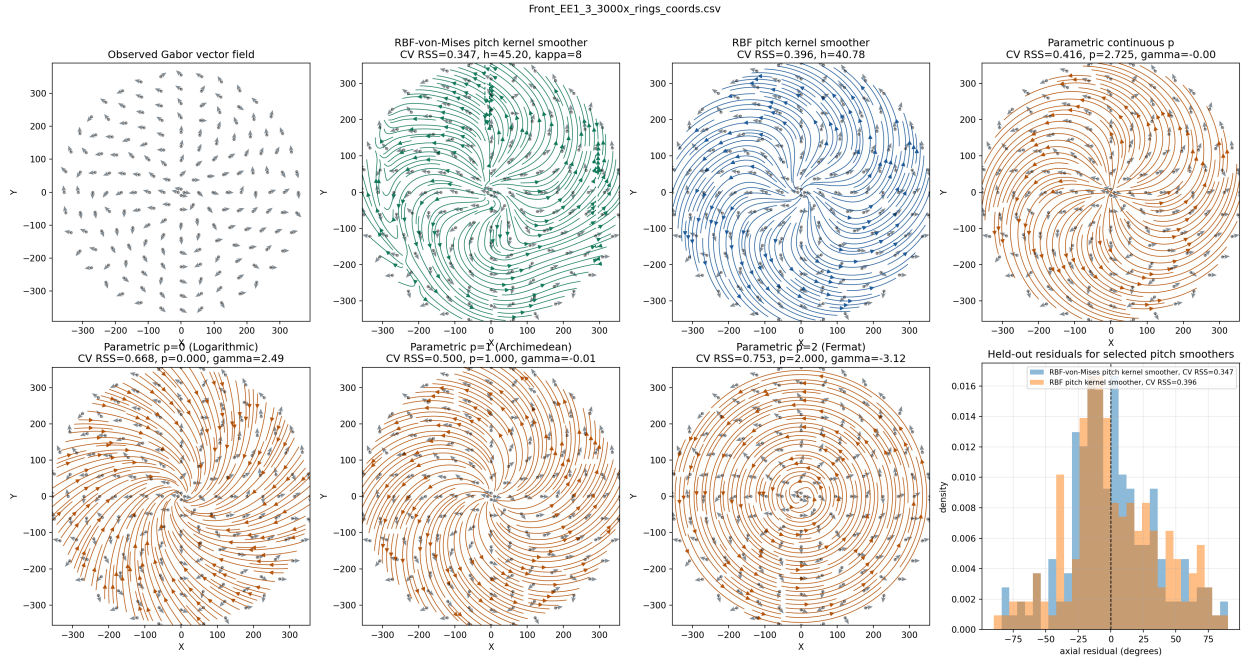


Figure 10: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE1_3_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

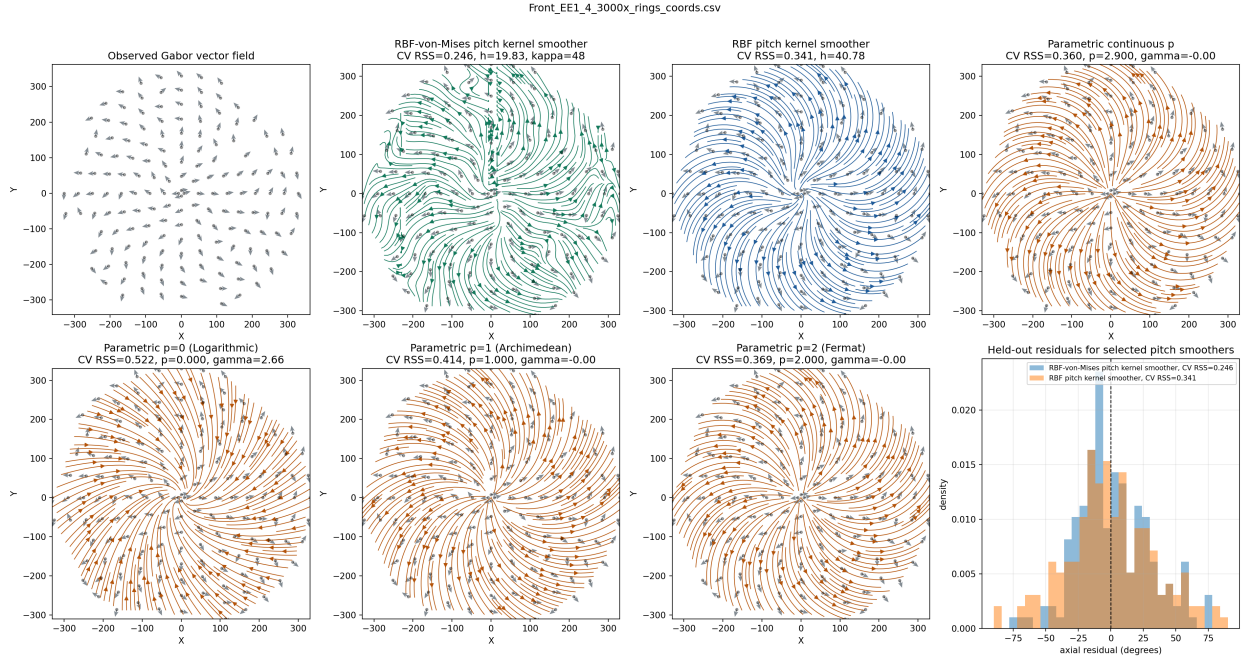


Figure 11: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE1_4_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.

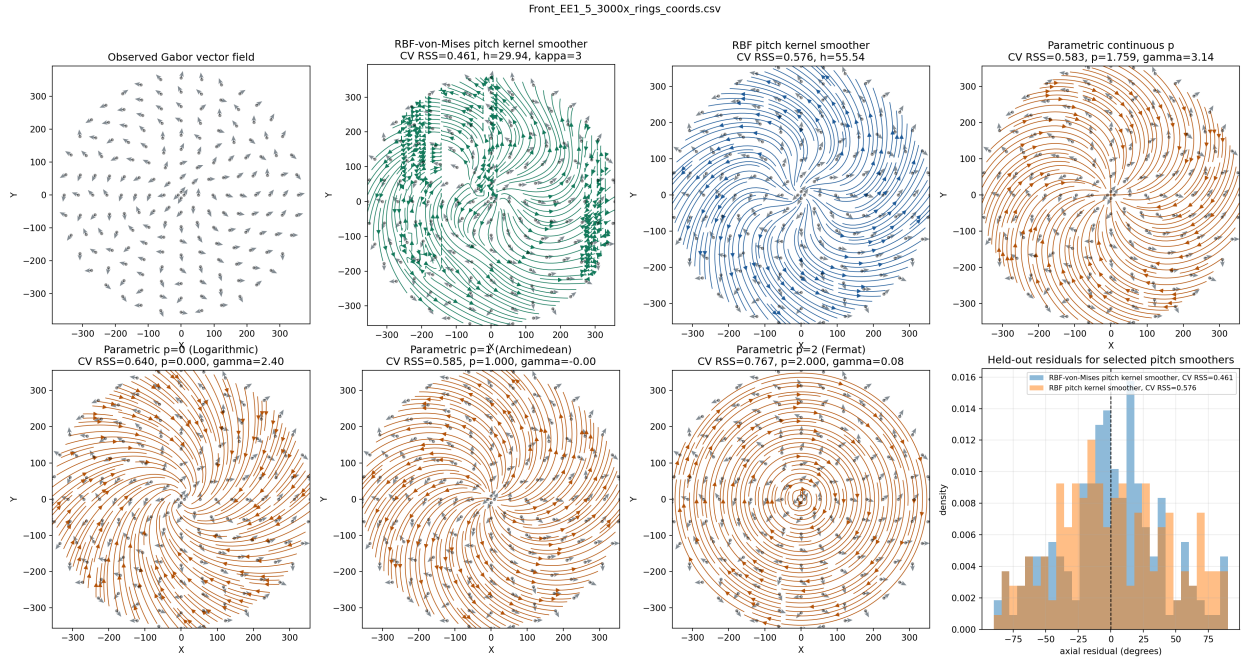


Figure 12: Notebook-style pitch-kernel and parametric streamline overlays for `Front_EE1_5_3000x_rings_coords.csv`. Observed Gabor arrows are shown at opacity $\alpha = 0.7$.